

C:\Users\mthob\OneDrive\Documents\NetBeansProjects\FederatedIDSArtefact\src\main\java\com\registration\app\federatedid

```
1 /*
2  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3  * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4  */
5 package com.registration.app.federatedidsartefact;
6
7 /**
8  *
9  * @author mthobi
10  * Aggregation - Only model parameters are aggregated; raw data never leaves the clients
11  * that spells out the data safety and security in this model training.
12  */
13
14
15 import java.util.*;
16
17 public class FederatedServer {
18
19     public static Model federatedAverage(List<Model> models) {
20         int numFeatures = models.get(0).weights.length;
21         Model globalModel = new Model(numFeatures);
22
23         for (Model m : models) {
24             for (int i = 0; i < numFeatures; i++) {
25                 globalModel.weights[i] += m.weights[i];
26             }
27             globalModel.bias += m.bias;
28         }
29
30         int n = models.size();
31         for (int i = 0; i < numFeatures; i++) {
32             globalModel.weights[i] /= n;
33         }
34         globalModel.bias /= n;
35
36         return globalModel;
37     }
38 }
39
40
```